ThermalSense: Autonomous UAV Patrol System

Name: **Md. Shakhoyat Rahman Shujon** | Roll: **2107104**

1. What is the Project About

This project presents an interactive 3D simulation of an autonomous drone patrolling a nighttime landscape, developed using OpenGL. Instead of standard rendering, the scene simulates a thermal imaging camera, displaying a localized "heat" color map. The project demonstrates fundamental computer graphics concepts including 3D transformations, multiple shading models (Flat, Gouraud, and Phong), depth testing for hidden surface removal, and algorithmic edge rendering. Three types of light sources will be used simultaneously to create dynamic illumination and visual heat effects.

2. Objects Used in the Project

- **Command Bunker:** A blocky, low-poly outpost structure that utilizes Flat Shading to emphasize its sharp architectural edges.
- **Terrain Map:** A rolling landscape serving as the environment, utilizing Gouraud Shading to compute lighting for a smooth, matte surface.
- **Autonomous UAV (Drone):** The main patrol vehicle flying overhead, utilizing Phong Shading for precise, moving specular highlights on its metallic chassis.
- **Target Vehicle (Rover):** A moving ground object that navigates the terrain to demonstrate dynamic depth interactions.
- **Light Sources:** Three types of lighting (Directional Lunar Ambient, Point Light Bunker Beacon, and Dynamic Spotlight UAV Searchlight) to illuminate the scene.

3. Features and Motions

The project will demonstrate translation, rotation, and scaling of 3D objects. The UAV drone will animate to fly across the sky, while the target rover will move along the ground terrain. The drone's attached searchlight will use hierarchical motion to dynamically cast a cone of light over the landscape. The project features three distinct lighting models interacting with three different surface rendering techniques to show how light behaves on different materials. It will use Z-buffer depth testing for accurate hidden surface removal (e.g., hiding the rover when it drives behind the bunker). Additionally, an interactive wireframe mode will display algorithmic edge rendering to visualize non-overlapping and overlapping edges. Camera controls will allow users to view the scene from different angles and toggle the diagnostic views.